§6.6 壁湍流相干结构

定义：

湍流的相干结构是指三维流场中的一个区域，在这个区域内至少有一个流动物理量（如速度分量、密度、温度等）在远大于当地流动最小尺度的时空范围内与自身或其它物理量显著相关。

特征：

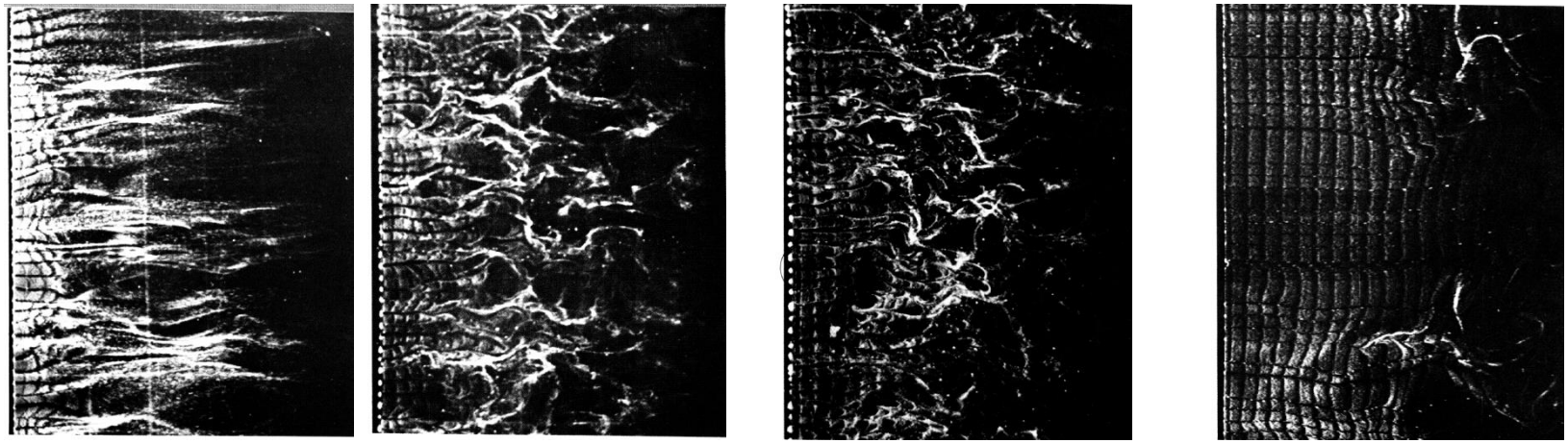
相干结构也称为拟序结构或拟序运动，是因为它是切变湍流中不规则触发的一种序列运动，它的起始时刻和位置是不确定的，但一经触发，它就以某种确定的次序发展特定的运动状态。

§6.6 壁湍流相干结构

- 壁湍流中的相干结构
  - 条带结构
  - 涡结构：流向涡，发卡涡
  - 猝发过程：上抛，下扫

§6.6 壁湍流相干结构

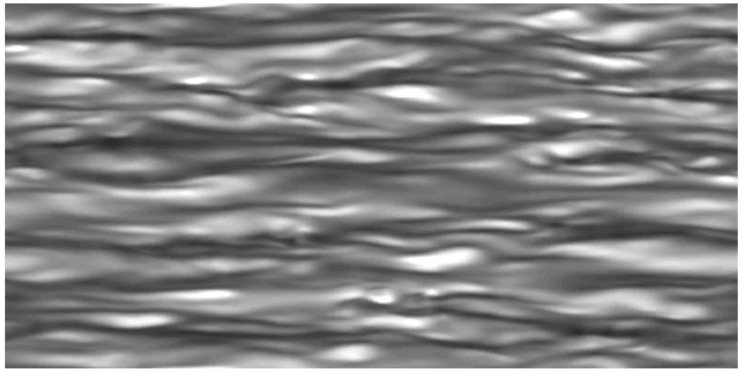
条带结构



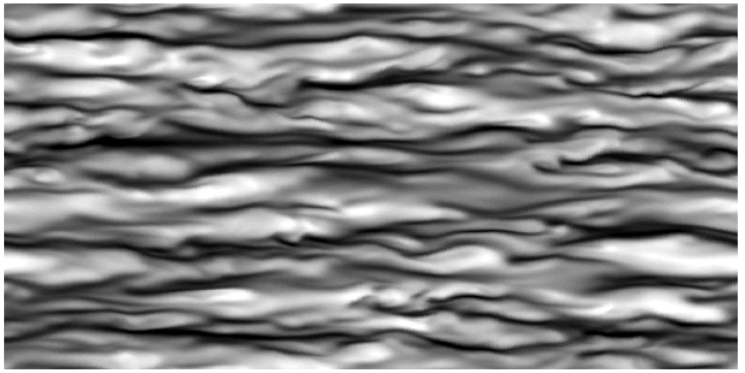
$y^+ = 2.7$　　$y^+ = 38$　　$y^+ = 101$　　$y^+ = 407$

氢气泡流动显示 (Kline 1967)

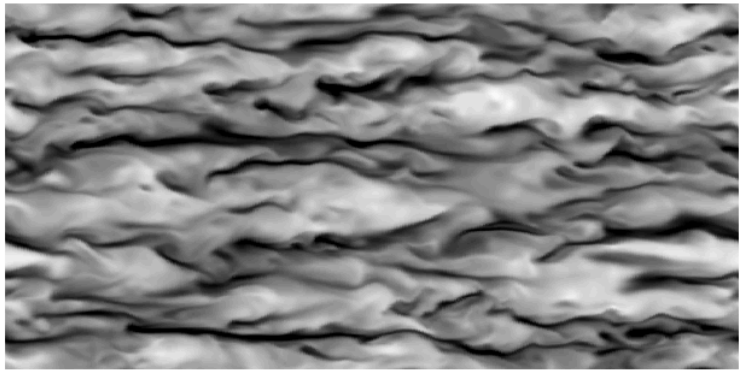
§6.6 壁湍流相干结构



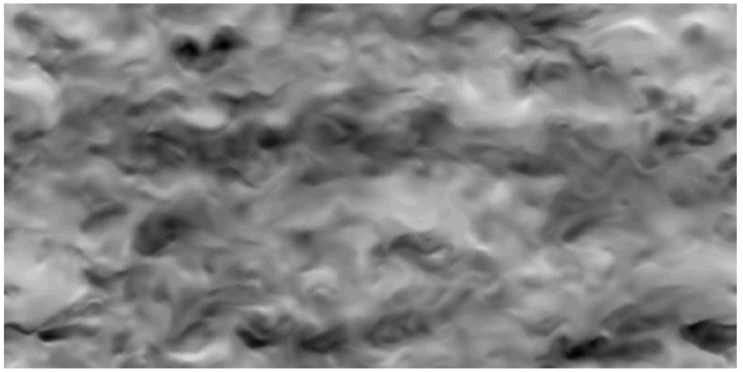
$y^+ = 3.4$



$y^+ = 12$



$y^+ = 30$



$y^+ = 102$

直接数值模拟

§6.6 壁湍流相干结构

近壁涡结构

涡识别方法

拟涡能（涡量）： $\frac{1}{2}|\boldsymbol{\omega}|^2$

低压判据： $p' < 0$

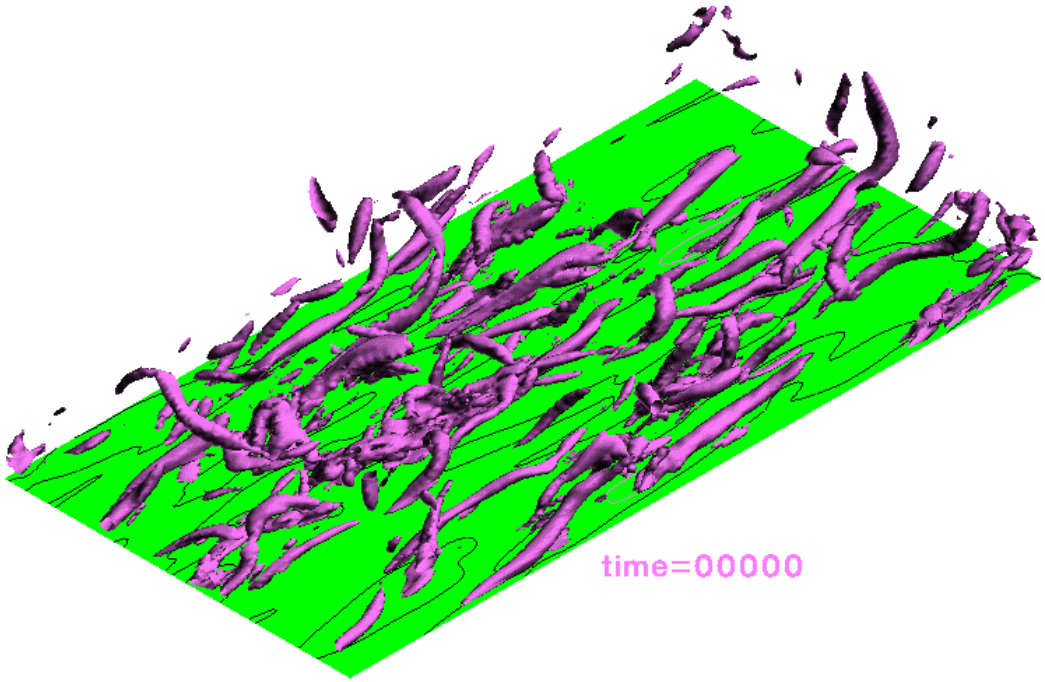
速度梯度张量：

$$Q = \frac{1}{2}\left(\Omega_{ij}\Omega_{ij} - S_{ij}S_{ij}\right) > 0$$

Δ

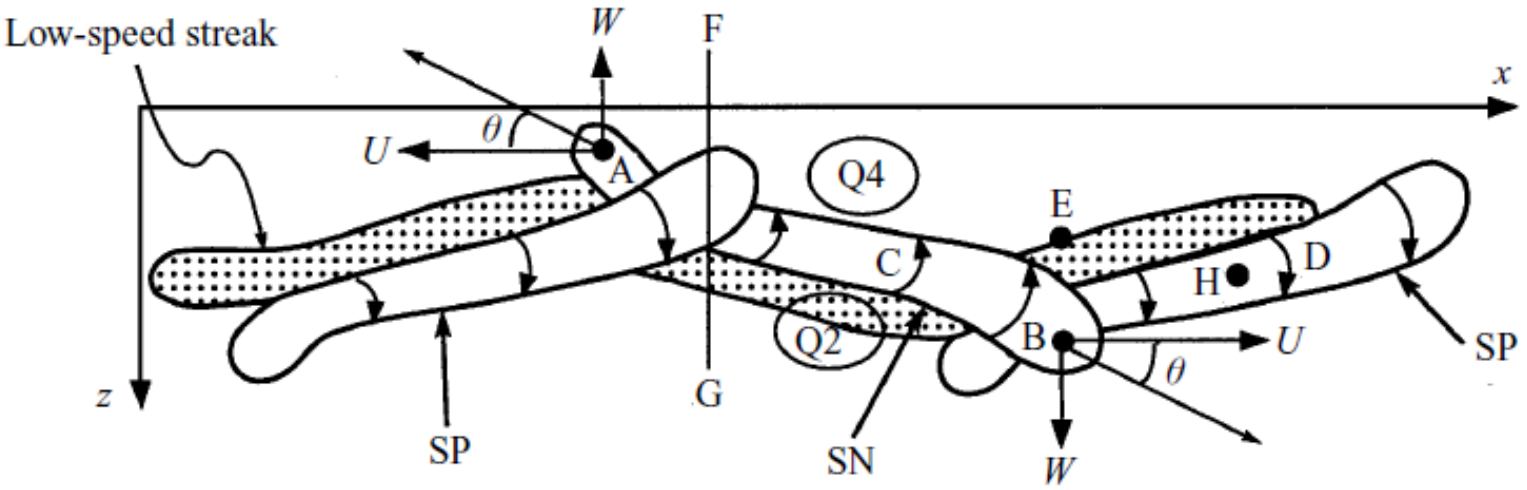
λ_2

(Jeong & Hussain 1995 JFM)



§6.6 壁湍流相干结构

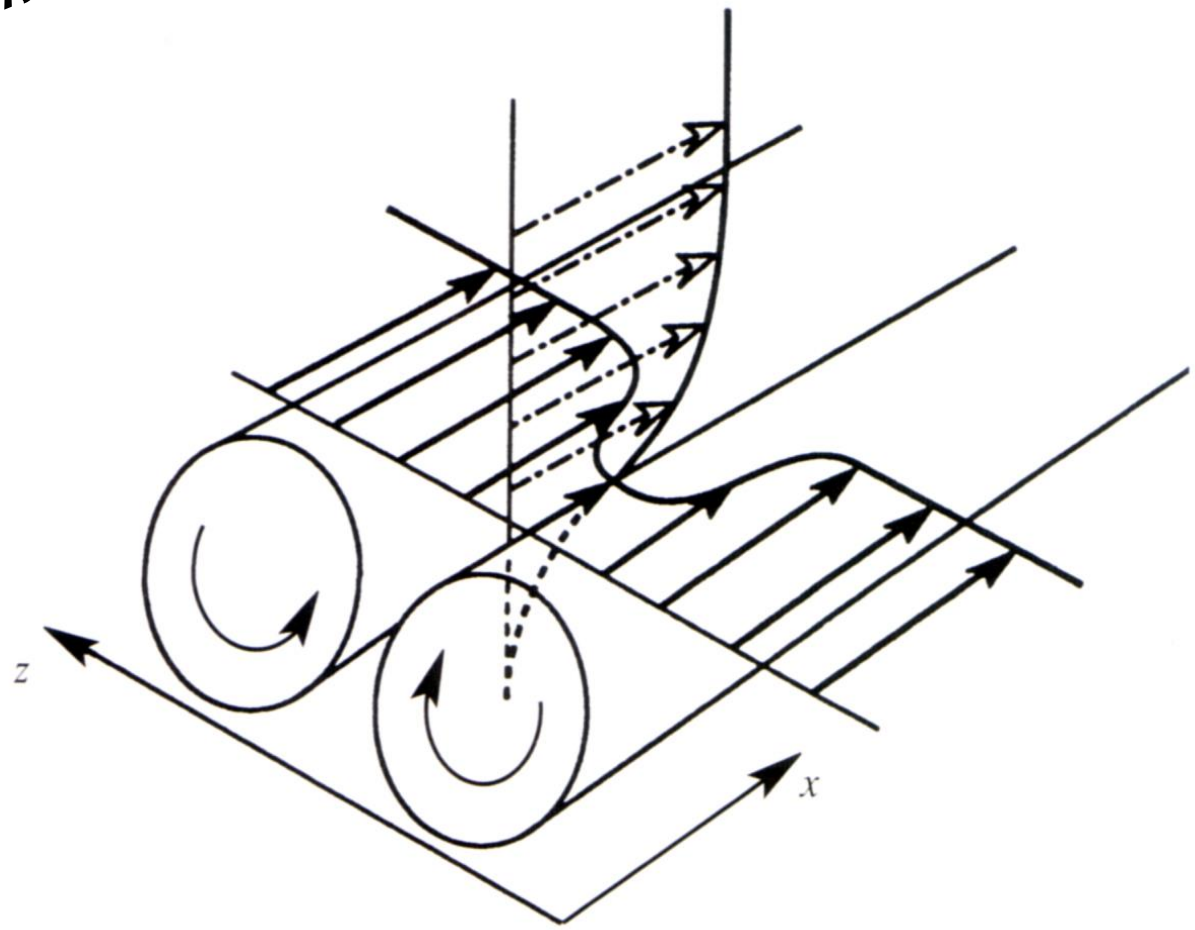
流向涡的空间分布



Jeong & Hussain, JFM,1997

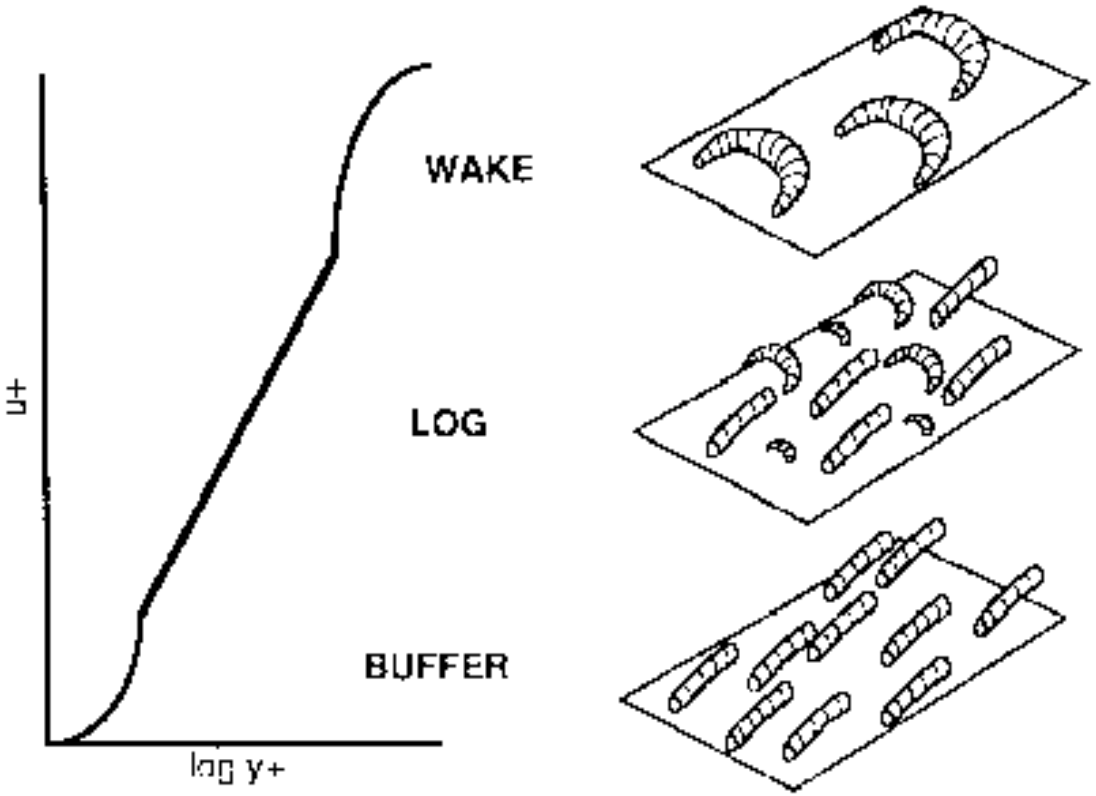
§6.6 壁湍流相干结构

涡与条带



§6.6 壁湍流相干结构

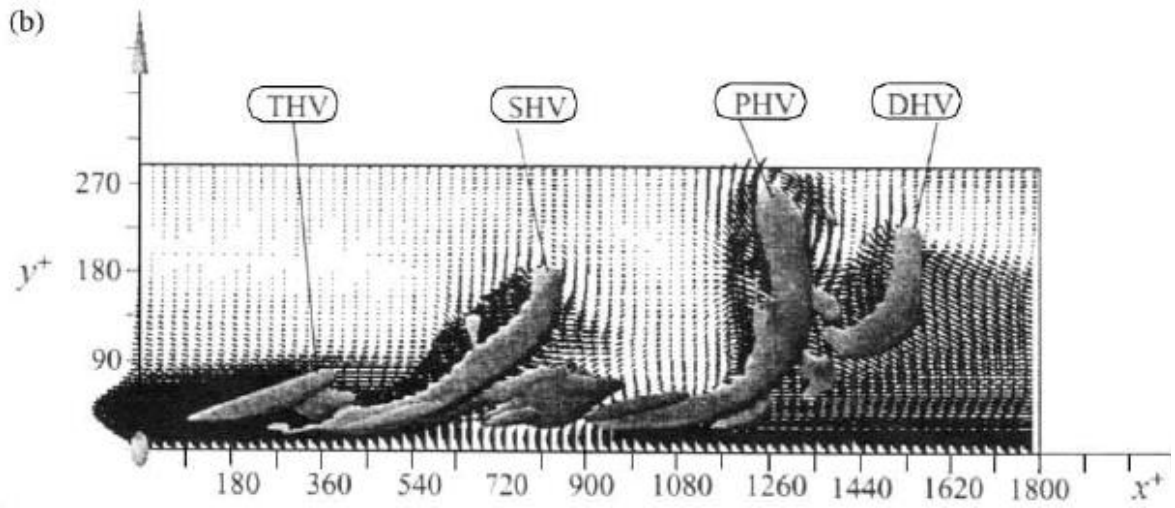
壁湍流中的发卡涡



Hairpin vortex defined by Robinson (1991), based on turbulent boundary layer DNS database of Spalart (1988)

§6.6 壁湍流相干结构

发卡涡包的形成

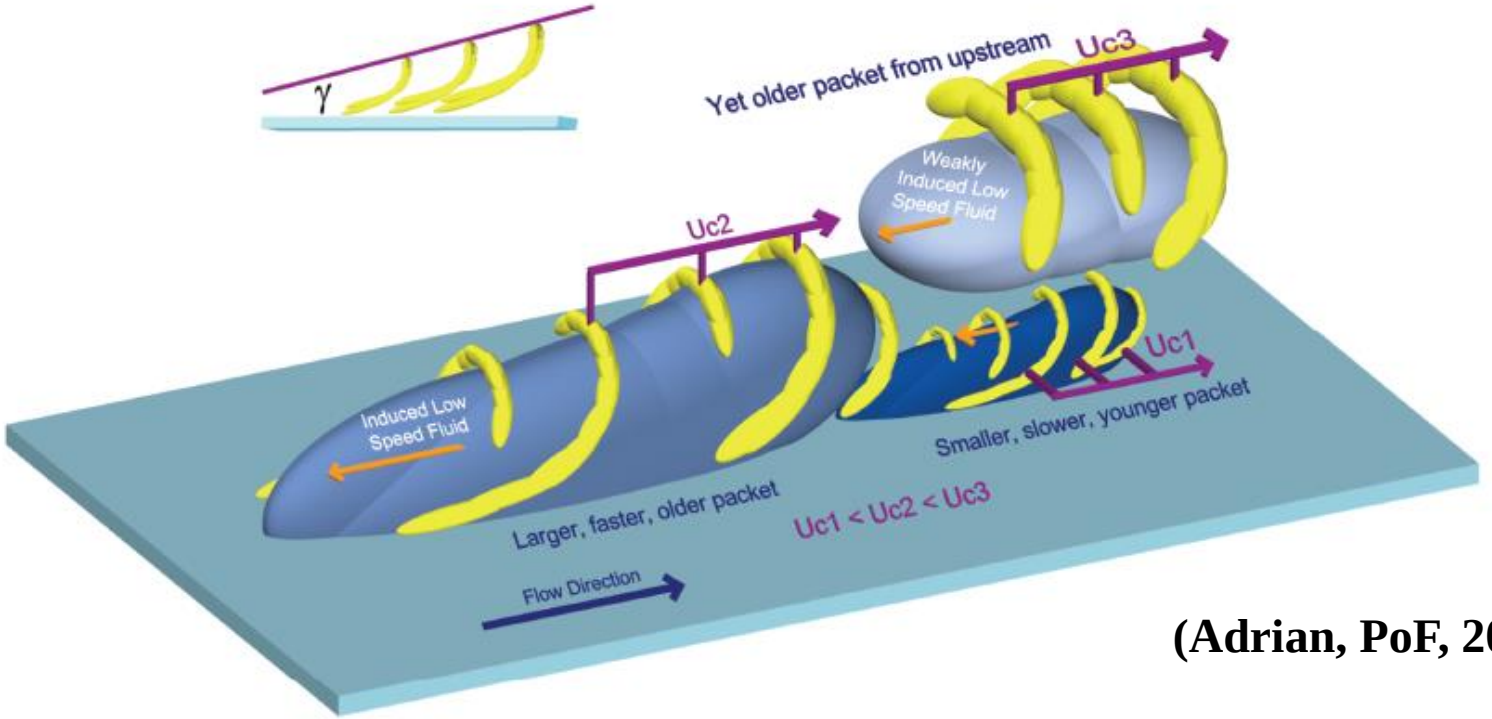




Zhou & Adrian, JFM, 1999
Adrian, J.Vision， 2002

§6.6 壁湍流相干结构

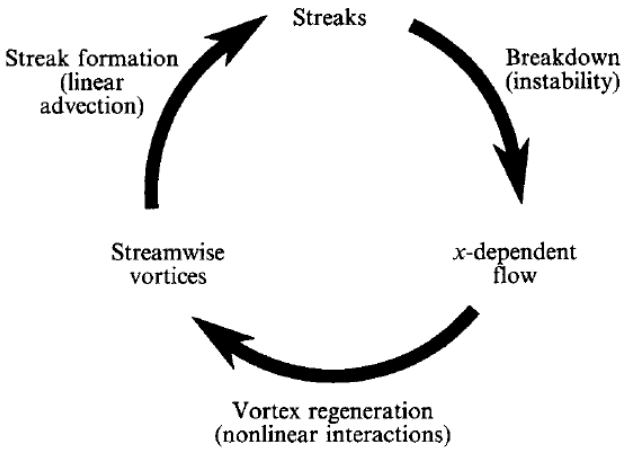
发卡涡涡包 (hairpin vortex packet)



(Adrian, PoF, 2007)

§6.6 壁湍流相干结构

近壁相干结构在湍流的维持中的重要地位



Hamilton et al. 1995 JFM

近壁湍流存在自维持过程

（Hamilton et al. 1995 JFM; Jimenez and Pinelli,1999, JFM）

- 流向涡生成近壁条带，条带破碎，流向涡再生成；
- 该循环过程不依赖外区流动；
- 人为打破任何一个过程会导致层流化；

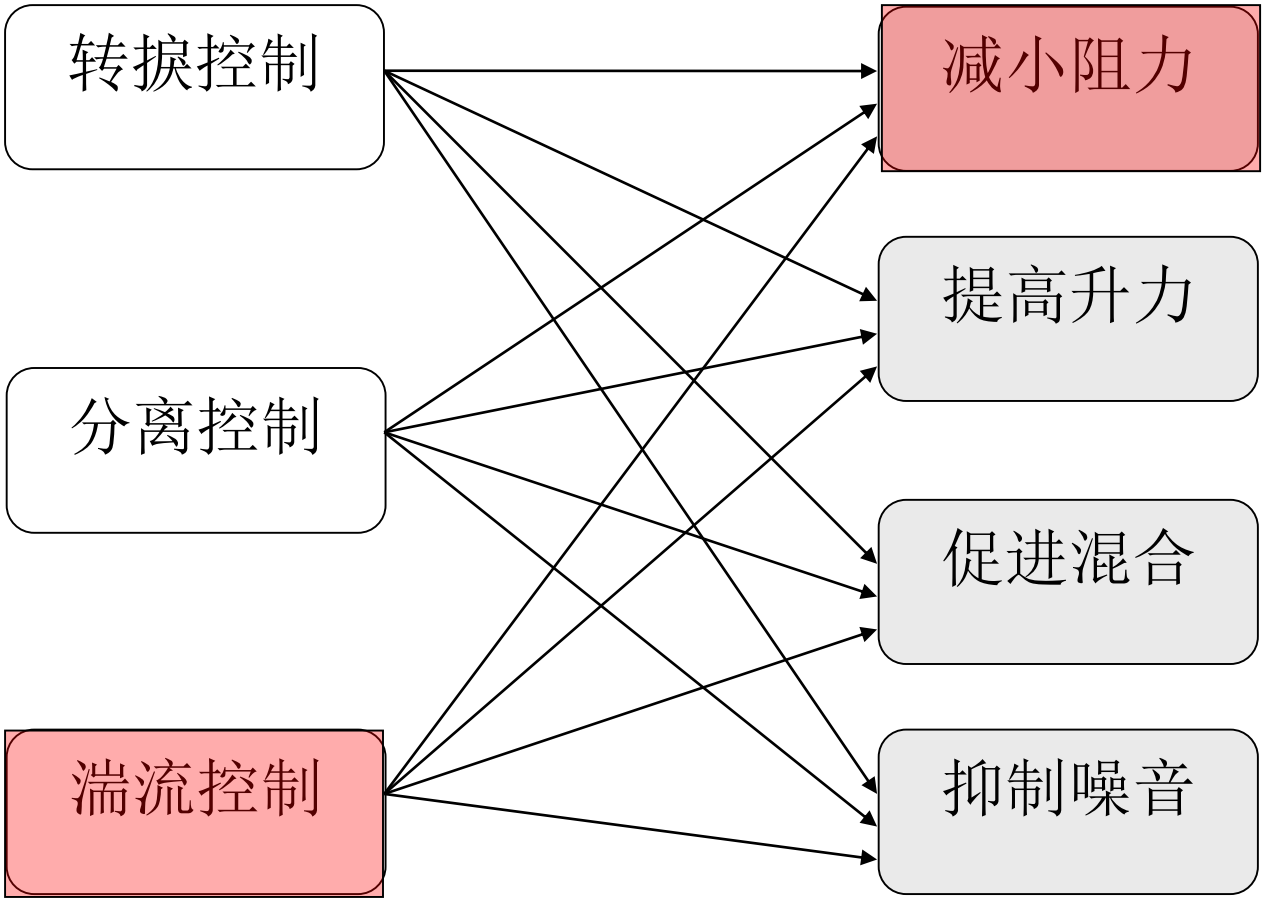
§6.6 壁湍流相干结构

- 流动控制

  流动控制是指通过改变流体运动的行为从而达到预期目的的机理或过程。

§6.6 壁湍流相干结构

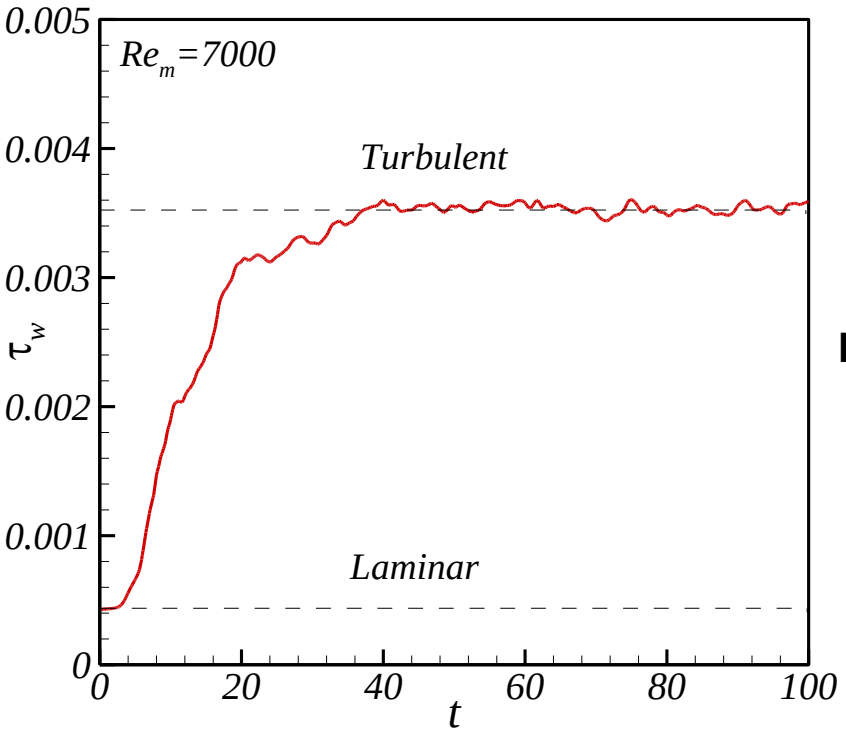
流动控制内容

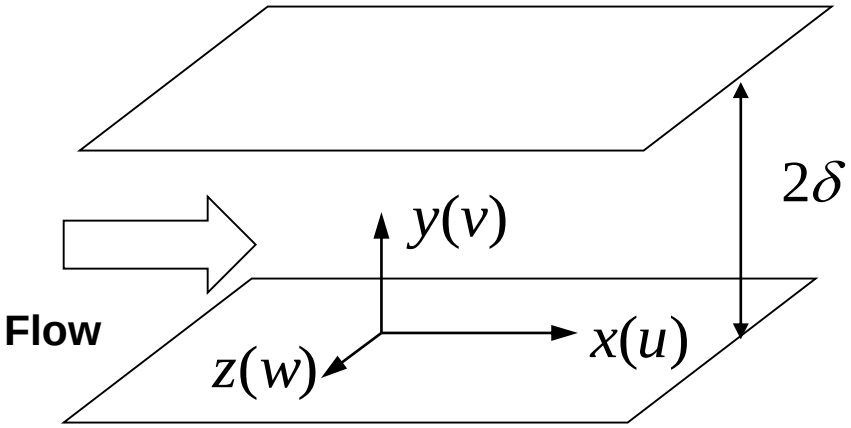


§6.6 壁湍流相干结构

- 湍流减阻控制

以槽道湍流为例介绍湍流摩擦阻力的来源



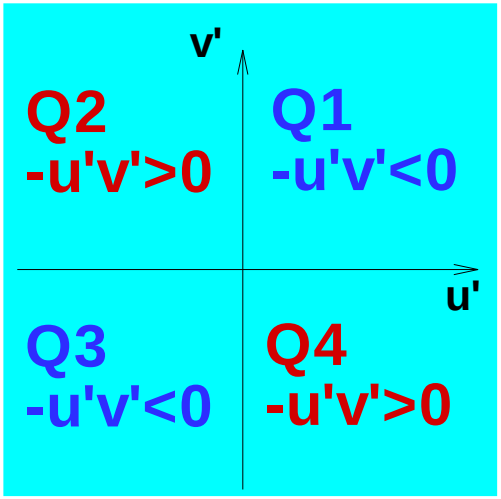


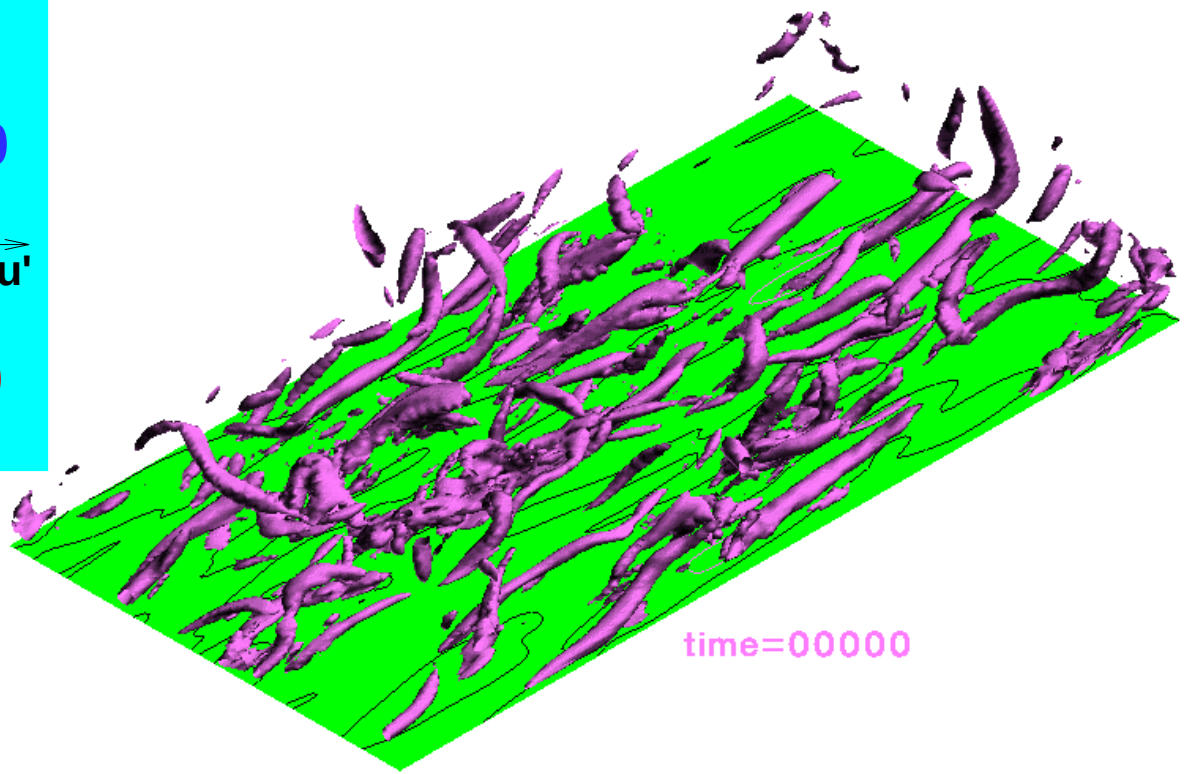
FIK公式：

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho U_m^2} = \frac{12}{\mathrm{Re}} + 24\int_0^1 (1-\eta)\langle -u'v' \rangle d\eta$$

§6.6 壁湍流相干结构

- $-\langle u'v' \rangle$ 的产生
- 流向涡：高速流体下扫，低速流体上抛





§6.6 壁湍流相干结构

- 湍流减阻控制方案
- 被动控制：

  外加物质、改变壁面几何形状

  优点：不需外加能量

  缺点：不适应流动变化
- 主动控制：

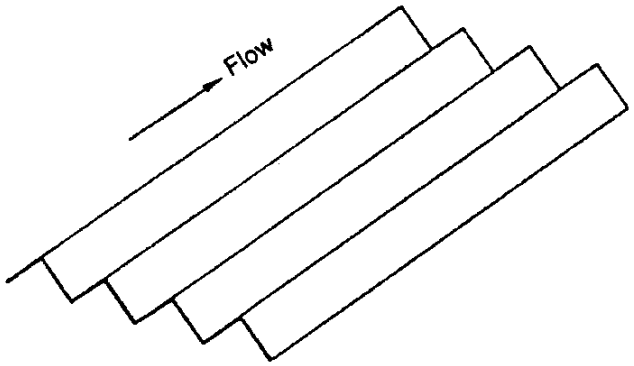
  壁面吹吸、展向壁面周期运动、电磁力

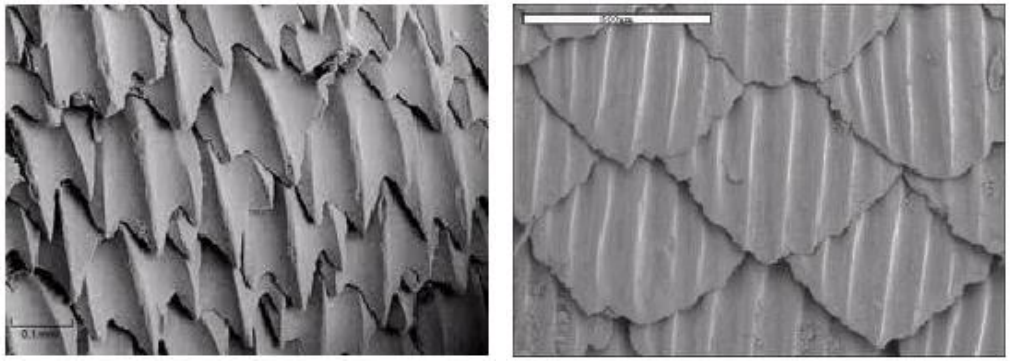
§6.6 壁湍流相干结构

- **改变壁面形状减阻**
  - **壁面沟槽(Riblets)**

    谷：减小40%

    峰：增加10%





鲨鱼：皮肤纵向小齿

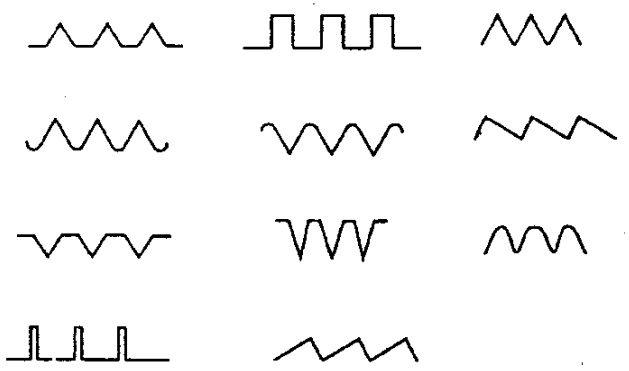


Figure 10.5: Fin shapes for longitudinally ribbed surface.

§6.6 壁湍流相干结构

- 改变壁面形状减阻
  - 柔性覆层



海豚：柔性皮肤

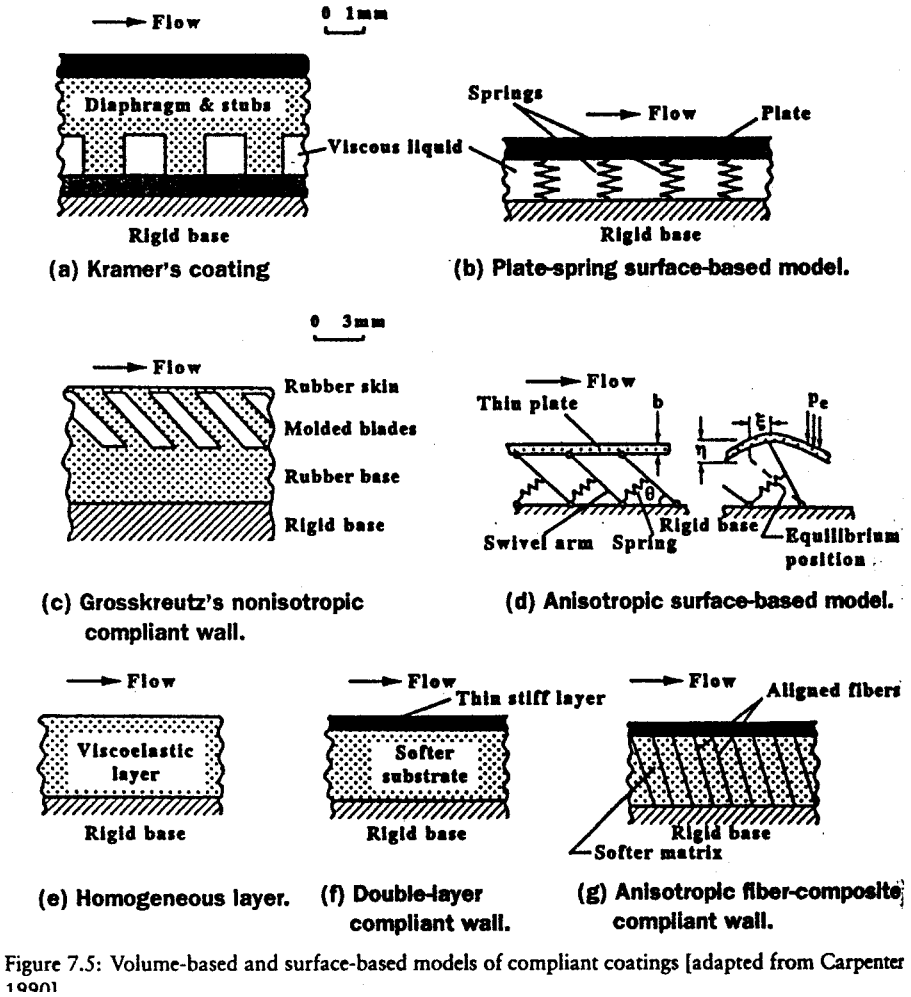
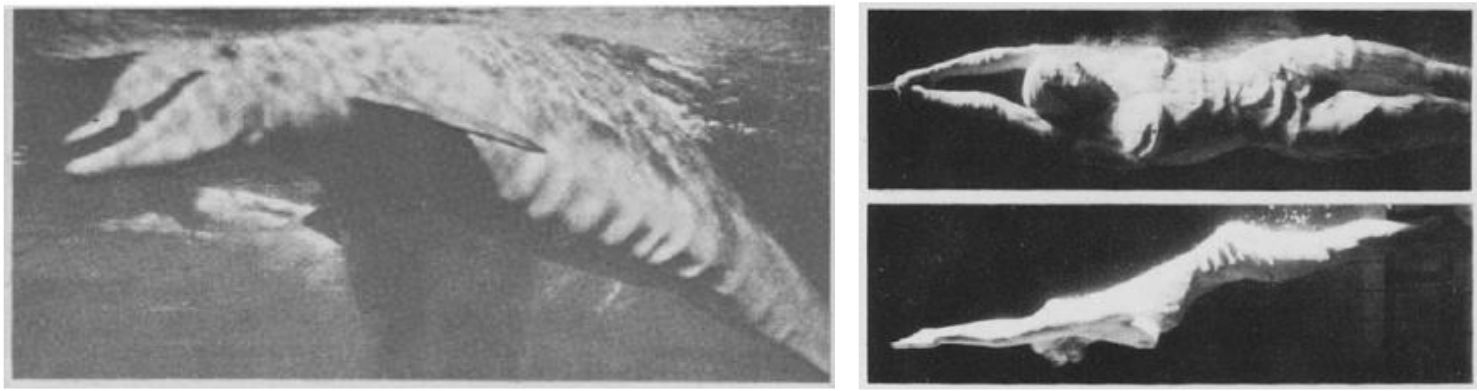


Figure 7.5: Volume-based and surface-based models of compliant coatings [adapted from Carpenter 1990]

§6.6 壁湍流相干结构

- 改变壁面形状减阻
  - 柔性覆层

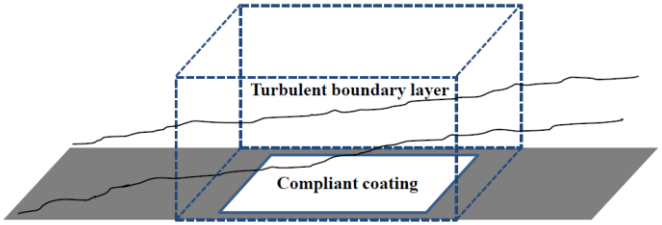


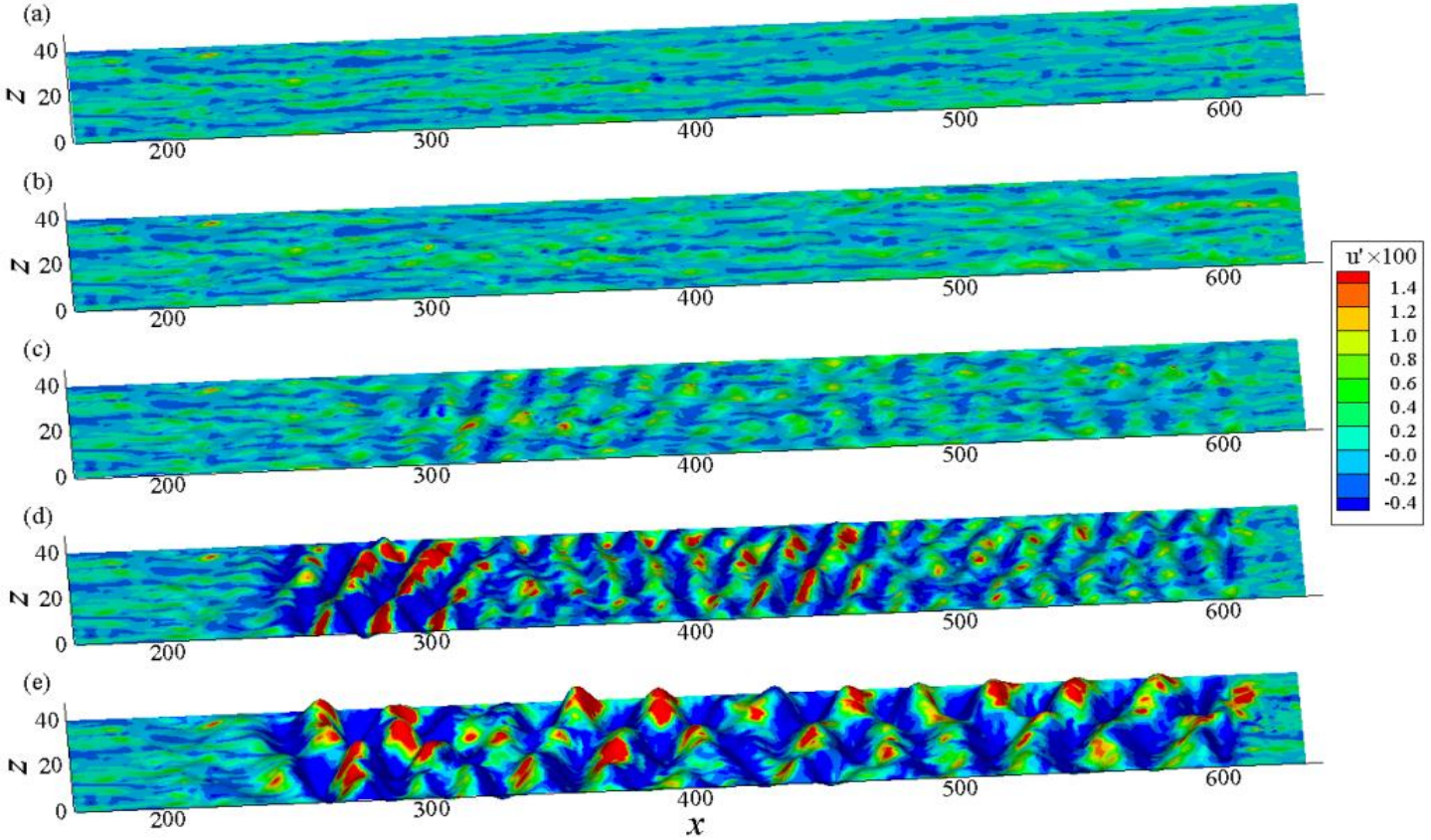
this allows the conclusion that in dolphins too moving transverse wavy skin folds form passively during movement, i.e. independently of the action of any muscles, and solely as a result of a certain relationship holding between the dynamic pressure of the flow and the elastic properties of the skin.

(Aleyev, Nekton, 1977)

§6.6 壁湍流相干结构

- 改变壁面形状减阻
  - 柔性覆层

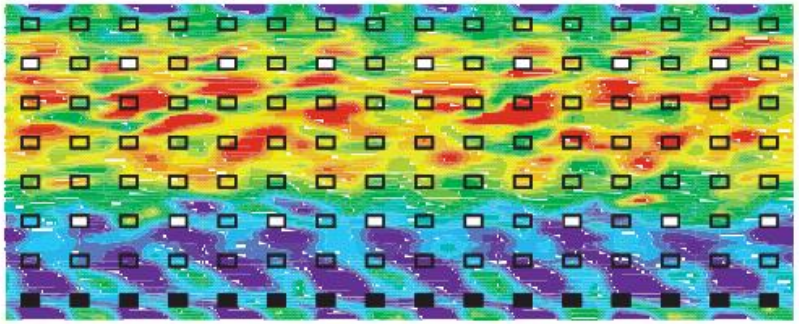




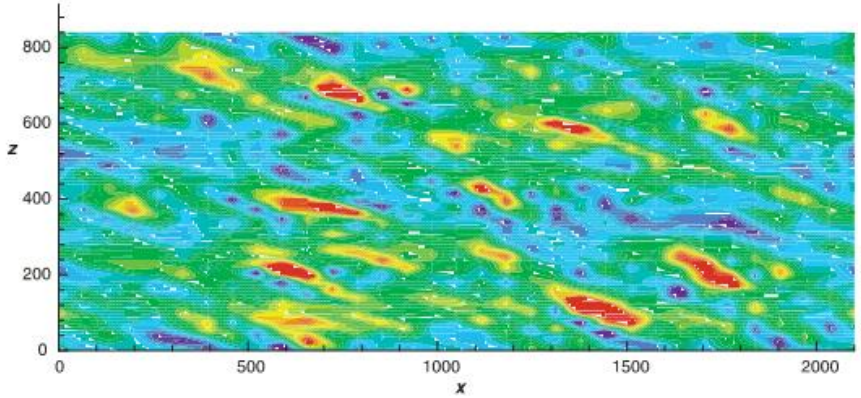
§6.6 壁湍流相干结构

展向振动减阻控制

展向振荡电磁力控制



展向速度分布



流向速度分布

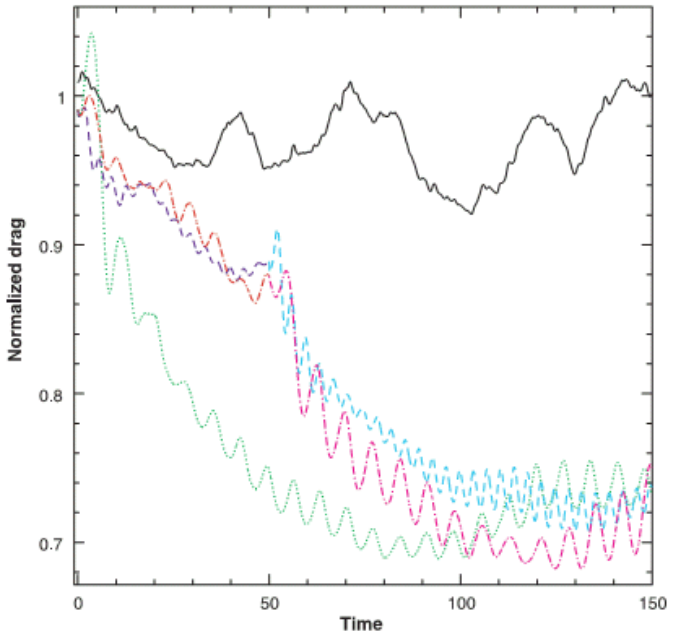


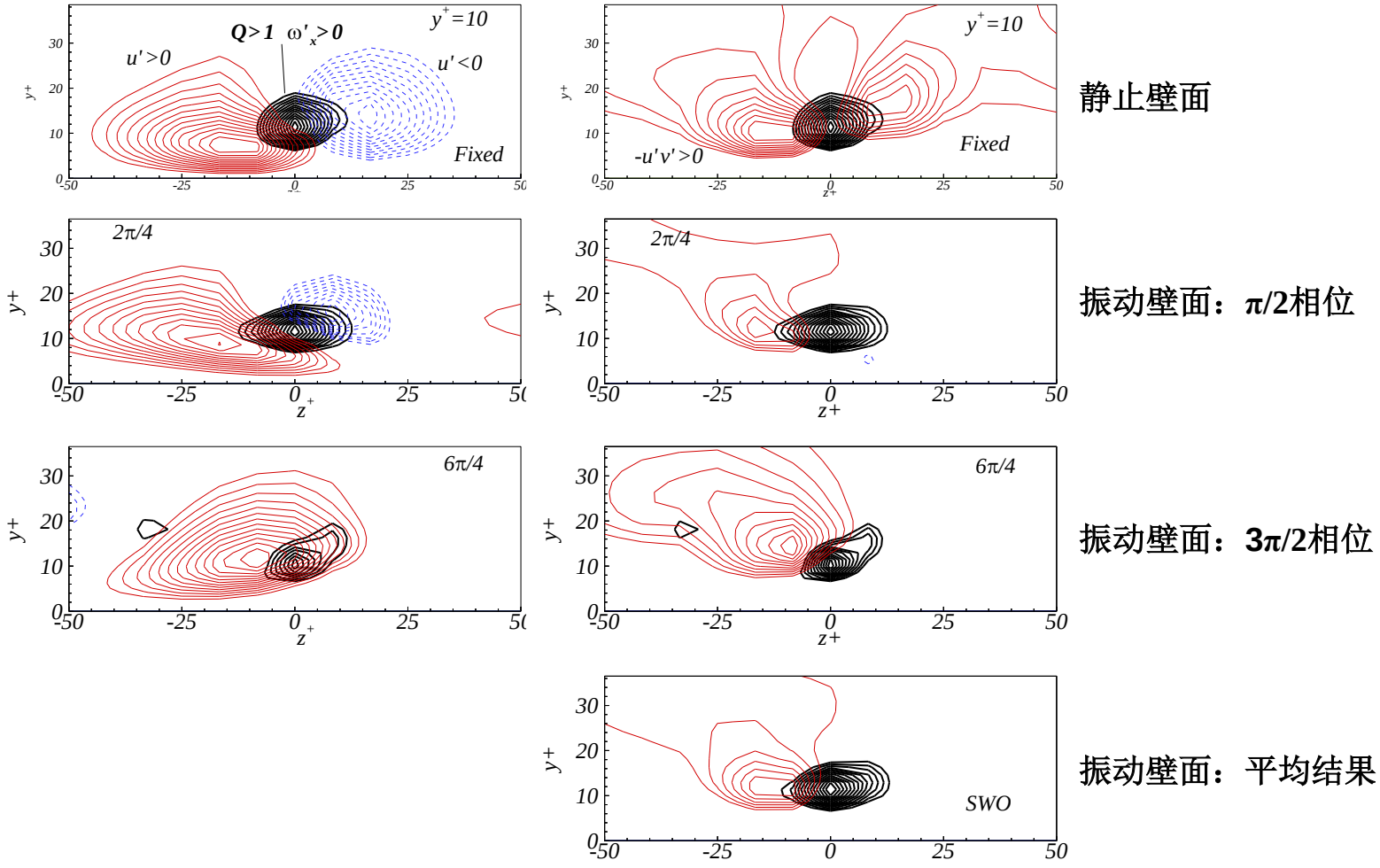
Fig. 5. Time history of normalized drag force for $\lambda_z^+ = 840$. The no-control value of the drag is used in the nondimensionalization, and time is in convective units. Black line corresponds to no control, green line to $l = 1$; $T^+ = 50$, red line to $l = 0.5$; $T^+ = 50$, pink line to $l = 1$; $T^+ = 50$, blue line to $l = 1$; $T^+ = 25$, and cyan line to $l = 2$; $T^+ = 25$.

Du & Karniadakis, Science, 2002

§6.6 壁湍流相干结构

展向壁面振动减阻控制

条件平均结果



静止壁面

振动壁面：π/2相位

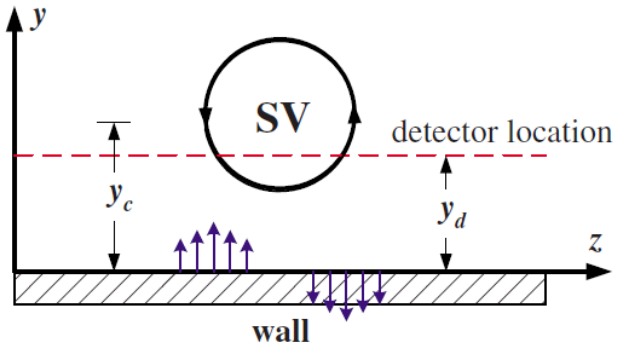
振动壁面：3π/2相位

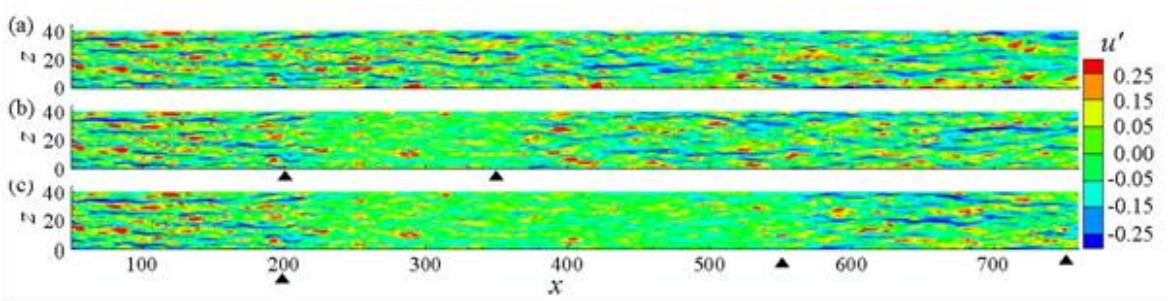
振动壁面：平均结果

§6.6 壁湍流相干结构

湍流边界层的反向控制

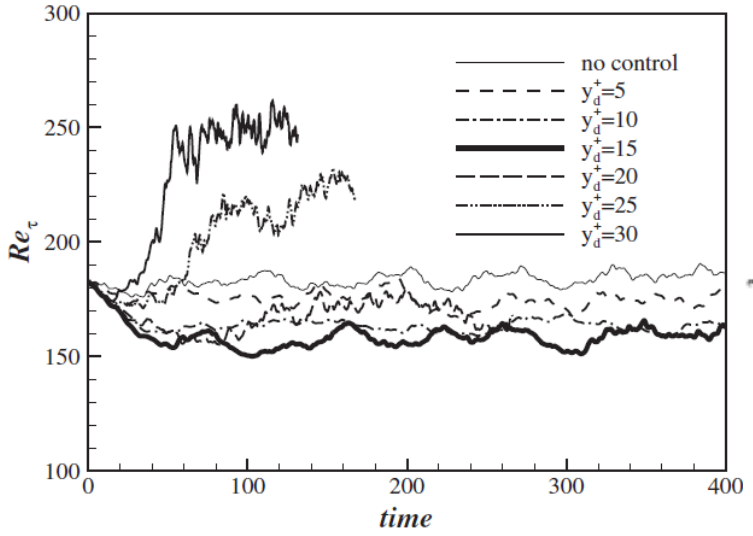
$$v(x,0,z;t) = -v(x,y_d,z;t)$$

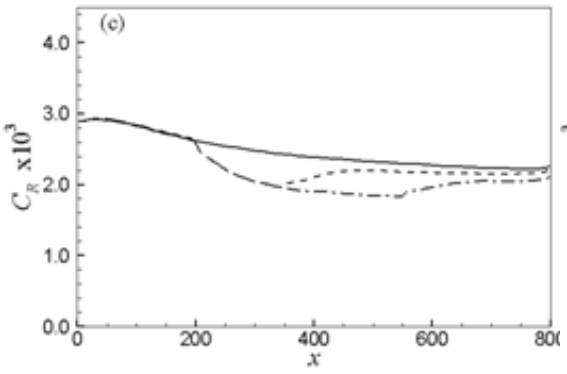


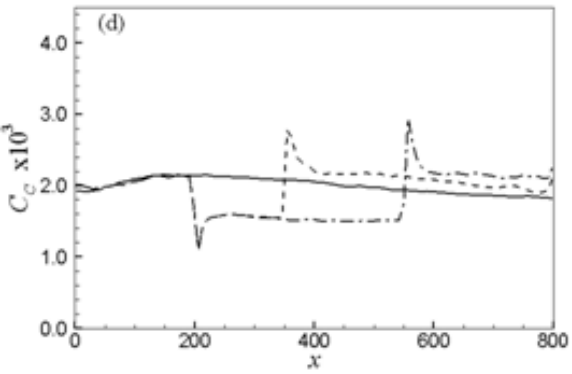


$$C_f = \underbrace{\frac{2}{\mathrm{Re}_\delta}}_{C_\upsilon(x)} + \underbrace{2\int_0^\delta \left(-\langle u'v'\rangle\right)dy}_{C_R} + \underbrace{2\int_0^\delta \left(1-\frac{y}{\delta}\right)\left(-\frac{\partial UU}{\partial x}-\frac{\partial UV}{\partial y}-\frac{\partial P}{\partial x}\right)d\left(\frac{y}{\delta}\right)}_{C_C}$$

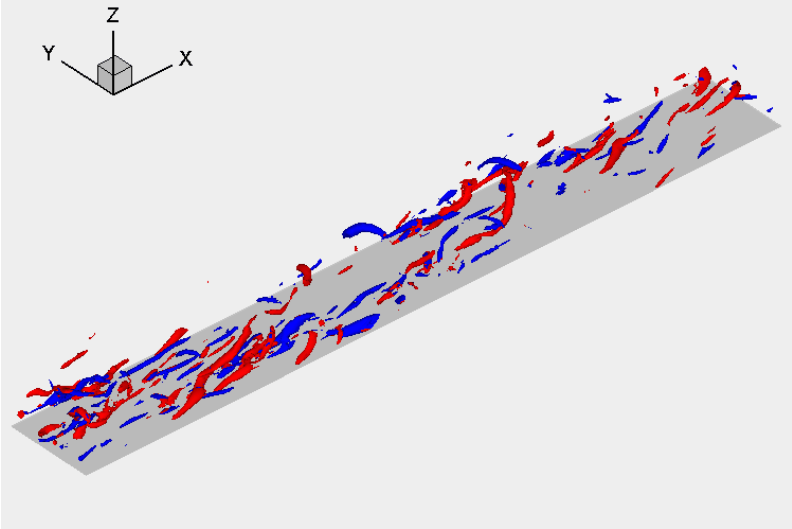
$$+\underbrace{2\int_0^\delta \left(1-\frac{y}{\delta}\right)\left(\frac{1}{\mathrm{Re}_\delta}\frac{\partial^2 U}{\partial x^2}-\frac{\partial\langle u'u'\rangle}{\partial x}\right)d\left(\frac{y}{\delta}\right)}_{C_D}$$



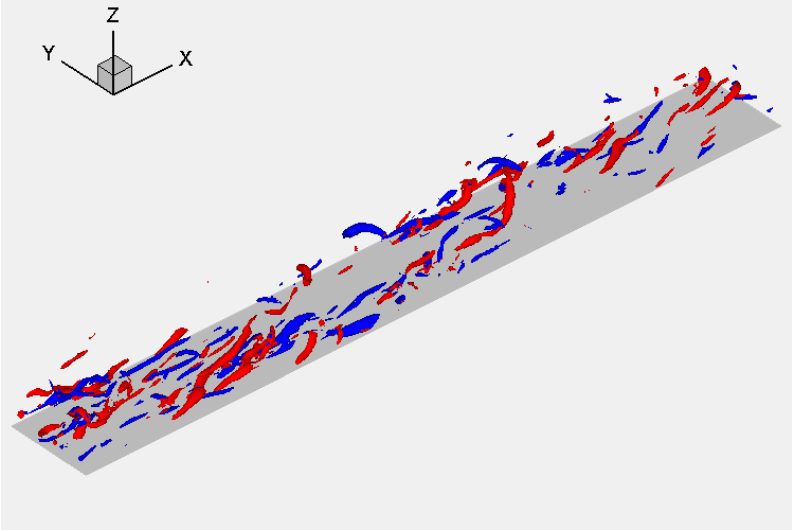


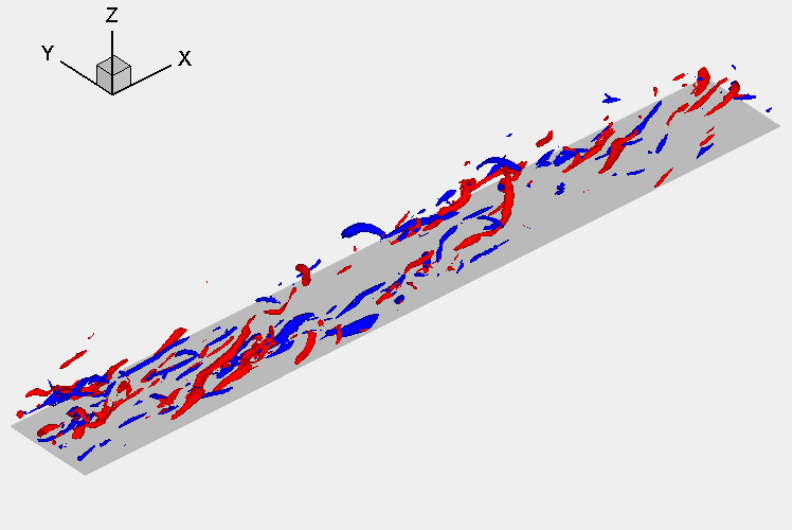


§6.6 壁湍流相干结构



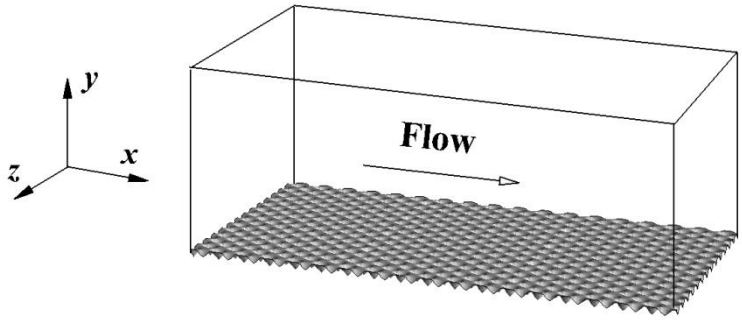
左上：无控制
左下：反向控制
右下：基于壁面变形的反向控制



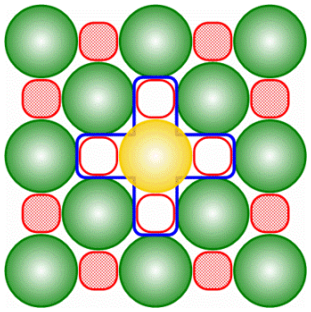


§6.6 壁湍流相干结构

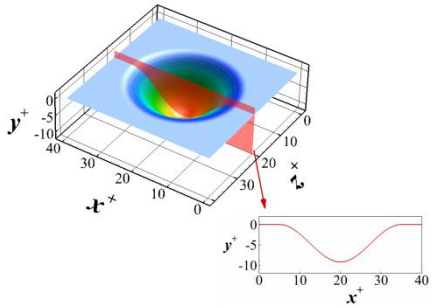
- 基于壁面凹坑变形的次优控制



流动示意图



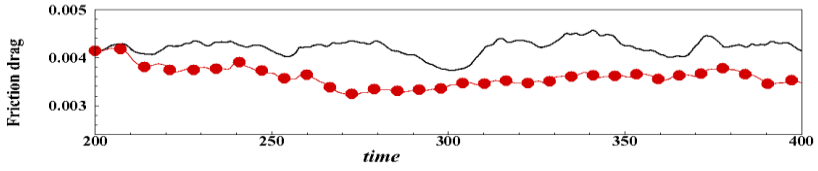
阵列传感器与凹坑

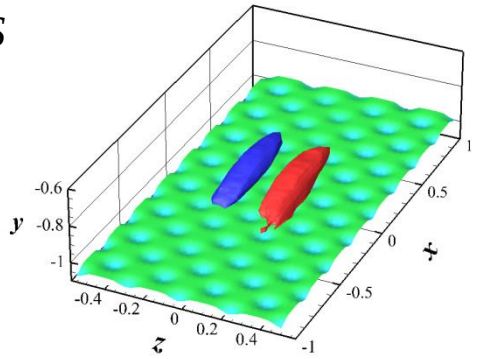


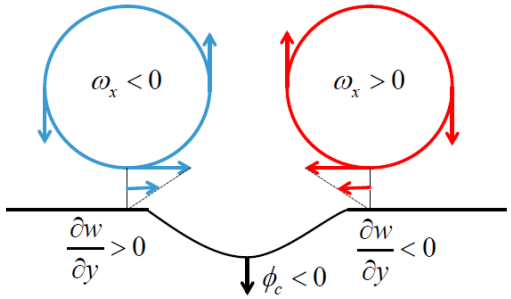
凹坑外形

$$\mathscr{J}(\phi)=\frac{1}{2A\Delta t}\int_S\int_t^{t+\Delta t}\left[l\phi^2-\left(\partial w/\partial y\right)_{\mathrm{w}}^2\right]\mathrm{d}t\mathrm{d}S$$

$$\phi\left(x_j,z_k\right)=C\sum_{j'}\sum_{k'}W_{j'k'}^{\mathrm{w}}\left.\frac{\partial w}{\partial y}\right|_{\mathrm{w}}\left(x_{j+j'},z_{k+k'}\right)$$







凹坑附近条件统计涡量场及控制机理示意图

本章结语

“流体是重要的又是难于理解的，关于Euler和Navier-Stokes方程解的行为有许多迷人的问题和猜想。因为，我们甚至不知道这些解是否存在，我们的理解还处于十分初步的水平。标准的偏微分方程的方法似乎不足以处理这种问题，作为替代，我们可能需要某些深刻的新的思想。”

—Charles Louis Fefferman （Fields Medal 1978）(1949-)